CS 210
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Program 3 Analysis

GITHUB LINK:

1. Briefly define a queue data structure.
A queue data structure follows a first in first out policy aka FIFO. Meaning the first input elements or whatever it is that goes into the queue first will also leave first and goes in that order. Think of it like a line at the SDSU market. If you get there first you get to leave first.

What evidence did you find that a queue container was operating in the load balancer?
Evidence to help support this would be requestQueue.front();
And requestQueue.pop().
This shows that it looks for it at the very front aka beginning and then pop removes the request so it constantly is getting the front first one then removing it exactly how a FIFO works.

2. Briefly define a stack data structure.
A stack data structure is a data structure that takes the last item that is placed in it and makes it the first to be removed. It's like a stack of plates when you're stacking the plates. The first one you place is at the bottom and is always the last to be grabbed because the newer plates that were placed get grabbed first. So the newer requests are dealt with first and the older ones last.

What evidence did you find that a stack container was operating in the load balancer?
This evidence is very similar to the first but the thing that changes is that it uses TOP() instead of front() while front looks at the front of it top() looks at the top of the data structure which is the newest one and then pop removes the request so it goes down the list of newest to oldest request.

Request r = requestqueue.top();
requestqueue.pop();

3. Briefly define a priority queue data structure.
A priority queue doesn't automatically remove anything; it follows rules which are determined by a comparator. There can be different case scenarios like for max processing time, min processing time or MAX PID.

What evidence did you find that a Max priority queue based on process time was operating in the load Balancer?

I could see that they were being used because of the outputs. Numbers like 362, 395, 399, and 480 were all popping up which shows the larger processing times which shows that it selected longer requests before the shorter ones meaning it would be a max priority que.

What evidence did you find that a Min priority queue based on process time was operating in the load
Balancer?

For this one the outputs were in ranges like 84, 87, 31,and 42 showing smaller processing times. This shows that the shorter times appeared first meaning that a min priority que was used because all the bigger times came last.

What evidence did you find that a Max priority queue based on process ID was operating in the load
Balancer?
This one was based off of the process ID being used. The first ID's popping up were in the range of 9-17 which showed that the PID's were higher and they were the newer requests. This means that a MAX priority queue was used because the Higher PID's were being used first.

Evidence can be a description of the output specific to the data structure and/or an annotated image.